which indicates that $\dot{\boldsymbol{\Theta}}\boldsymbol{\Theta}^T$ is a skew symmetric matrix. Any 3×3 skew matrix may be written as

$$\dot{\boldsymbol{\Theta}}\boldsymbol{\Theta}^T = \text{skew}(\boldsymbol{\omega}) \equiv \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}. \tag{4.38}$$

$\boldsymbol{\omega}$ is referred to as the *spatial angular velocity* of the rigid body. It follows that

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\delta}_i = \text{skew}(\boldsymbol{\omega})\boldsymbol{\delta}_i = \boldsymbol{\omega} \times \boldsymbol{\delta}_i = -\text{skew}(\boldsymbol{\delta}_i)\boldsymbol{\omega}, \tag{4.39}$$

which is the usual definition.

We now insert the expression for $\dot{\boldsymbol{\delta}}_i$ into the rotational kinetic energy, to obtain

$$K_{\text{rot}} = \frac{1}{2}\sum_{i=1}^{\text{N}} m_i \|\dot{\boldsymbol{\delta}}_i\|^2 = -\frac{1}{2}\sum_{i=1}^{\text{N}} m_i \boldsymbol{\omega}^T \text{skew}(\boldsymbol{\delta}_i)^2 \boldsymbol{\omega}.$$

It is convenient to express this in terms of the so-called *body angular velocity* $\boldsymbol{\omega}_{\text{b}} = \boldsymbol{\Theta}^T\boldsymbol{\omega}$ in which case it becomes

$$K_{\text{rot}} = \frac{1}{2}\boldsymbol{\omega}_{\text{b}}^T \boldsymbol{T} \boldsymbol{\omega}_{\text{b}},$$

where the matrix $\boldsymbol{T} = -\sum_{i=1}^{\text{N}} m_i \text{skew}(\boldsymbol{\delta}_i(0))^2$ is referred to as the *moment of inertia tensor* or simply the *inertia tensor* and can be written as

$$\boldsymbol{T} = \sum_{i=1}^{\text{N}} m_i \left[\|\boldsymbol{\delta}_i(0)\|^2 \boldsymbol{I} - \boldsymbol{\delta}_i(0)\boldsymbol{\delta}_i(0)^T \right].$$

This matrix is closely related to the matrix $\boldsymbol{R}$:

$$\boldsymbol{R} = \sum_{i=1}^{\text{N}} m_i \boldsymbol{\delta}_i \boldsymbol{\delta}_i^T = \sum_{i=1}^{\text{N}} m_i \|\delta_i\|^2 \boldsymbol{I} - \boldsymbol{T} = (T_{11} + T_{22} + T_{33})\boldsymbol{I} - \boldsymbol{T}. \tag{4.40}$$

Example 4.1 (Dipolar Molecular Liquid) The Stockmayer model for a molecular liquid with permanent dipoles [388] treats this as a system of rigid bodies (spheres) with dipole-dipole interaction potential; we already encountered this in the exercises of Chap. 1. From a computational and modelling perspective, there is no real reason to use spheres (and perhaps good reasons not to); for example they might be taken to be ellipsoidal. Let us assume that they are identical ellipsoids. The Hamiltonian for a more general dipolar system can be written

$$\begin{aligned} H = {} & K_{\text{rot}}(\boldsymbol{\pi}_1, \boldsymbol{\pi}_2, \ldots, \boldsymbol{\pi}_{\text{N}}) + K_{\text{trans}}(\boldsymbol{p}_1, \boldsymbol{p}_2, \ldots, \boldsymbol{p}_{\text{N}}) \\ & + U(\boldsymbol{q}_1, \boldsymbol{q}_2, \ldots, \boldsymbol{q}_{\text{N}}, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \ldots, \boldsymbol{\mu}_{\text{N}}), \end{aligned}$$

Where K_{rot} represents the rotational kinetic energy expressed in terms of the body-fixed angular momentum vectors, K_{trans} is the translational kinetic energy, U is the potential energy which is a function of all the positions and dipole orientations. The vectors $\boldsymbol{\mu}_k$ can be taken to be unit vectors reflecting the dipole orientation, so they are subject to the constraint $\|\boldsymbol{\mu}_k\| \equiv 1$. The total dipole-dipole interaction potential then takes the form

$$\varphi_{ij}^{\text{dip}} = \varphi_{\text{LJ}}(r_{ij}) + \boldsymbol{\mu}_i^T \boldsymbol{L}(\boldsymbol{q}_{ij})\boldsymbol{\mu}_j,$$

with U given as the sum of these over pairs. The 3×3 matrix $\boldsymbol{L}$ is given by

$$\boldsymbol{L}(\boldsymbol{q}_{ij}) = \frac{1}{r_{ij}^3} - \frac{3}{r_{ij}^5}\boldsymbol{q}_{ij}\boldsymbol{q}_{ij}^T, \qquad \boldsymbol{q}_{ij} = \boldsymbol{q}_i - \boldsymbol{q}_j.$$

The rotational kinetic energy may be written

$$K_{\text{rot}}(\boldsymbol{\pi}_1, \boldsymbol{\pi}_2, \ldots, \boldsymbol{\pi}_{\text{N}}) = \frac{1}{2}\sum_{i=1}^{\text{N}} \frac{\boldsymbol{\pi}_{i,1}^2}{I_1} + \frac{\boldsymbol{\pi}_{i,2}^2}{I_2} + \frac{\boldsymbol{\pi}_{i,3}^2}{I_3},$$

with I_1, I_2, I_3 the moments of inertia, taken to be diagonal with respect to a suitable coordinate system. Introducing a rotation matrix $\boldsymbol{\Theta}_k$ for the kth dipole, we have $\boldsymbol{\mu}_k(t) = \boldsymbol{\Theta}_k\boldsymbol{\mu}_k(0)$, where $\boldsymbol{\mu}_k(0)$ is a reference configuration of the dipole (its initial state). Then

$$K_{\text{rot}} = \sum_{i=1}^{\text{N}} \text{Tr}\left(\dot{\boldsymbol{\Theta}}_i \boldsymbol{R} \dot{\boldsymbol{\Theta}}_i^T\right),$$

with $\boldsymbol{R}$ defined in terms of $\boldsymbol{T} = \text{diag}(I_1, I_2, I_3)$ by (4.40).

We now have two alternatives for representing the rotational motion. In the first, the rotation matrix describing the rigid body's orientation is the dynamical quantity (and we need to enforce the orthogonality of this matrix as a constraint). This leads to a constrained integration method, which makes use of SHAKE (in a somewhat different way from the method [84] which constrains the relative positions of particles). In the second formulation, the orthogonality is used to perform a reduction to an angular velocity (or momentum) representation with respect to the body frame of reference.

A Hamiltonian description of the motion may be formulated in either system of notation. The Hamiltonian formulation with rotation matrices involves variables $\boldsymbol{q}_{\text{cm}}, \boldsymbol{p}_{\text{cm}}$ (phase space variables describing the center of mass motion) and $\boldsymbol{\Theta}, \boldsymbol{\Pi}$, which are 3×3 matrices. The Hamiltonian is (for a single rigid body):

$$H_{\text{RB}} = \frac{\|\boldsymbol{p}_{\text{cm}}\|^2}{2M} + \frac{\text{Tr}\left(\boldsymbol{\Pi}\boldsymbol{R}^{-1}\boldsymbol{\Pi}^T\right)}{2} + U(\boldsymbol{q}_{\text{cm}}, \boldsymbol{\Theta}),$$